

Technical Assessment: Non-Oracle Database Schema Enumeration

Assessor: Moe Htet Ar Kar (Phoe Cho) **Date:** July 2026

1. Vulnerability Identified

- **SQL Injection Schema Enumeration** (CWE-89 / CWE-200)

2. Proof of Concept (PoC)

1. Validated a SQL Injection flaw on a non-Oracle database (PostgreSQL/MySQL environment) where database types were initially unknown.
2. Exploited the flaw to query the built-in metadata tables (`information_schema`) and mapped out the entire database layout.
3. Extracted existing table names via: ' UNION SELECT table_name, NULL FROM information_schema.tables-- (identified target table: users_jwayxo).
4. Enumerated specific target column names via: ' UNION SELECT column_name, NULL FROM information_schema.columns WHERE table_name='users_jwayxo'-- (identified columns: username_lrcggf and password_fzhwkm).

3. Impact

Complete database structure disclosure under a black-box scenario, allowing attackers to target precise high-value authentication tables systematically.

4. Remediation

- Apply proper **Prepared Statements** to eliminate injection vectors.
- Restrict database user account privileges; ensure the application database user does not have unnecessary read access to the global `information_schema` views.